Forecasting of Renewable Energy in East England Using Machine Learning Models and Multi-Model Ensemble

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***Abstract*—Forecasting renewable energy (RES) generation is important for applications within power grid management, energy markets, operational planning and more. However, the uncertain nature of RES make it difficult to produce accurate forecasts. This project is based on an IEEE competition regarding forecasts of the combined wind and solar generation of Hornsea 1 and the solar capacity of East England. The aim is to produce forecasts that compete the results of previous competitors.**

Forecasts are produced using different machine learning algorithms and training them on data from the generation sites. The models used are Random Forest Regression (RFR), Light Gradient Boosting Machine (LightGBM), Categorical Boosting (CatBoost) and an ensemble model that produces a weighted combination of the predictions of these models. The forecasts are then evaluated using different error measures.

The results show that the model with the strongest performance is the ensemble method. This is consistent with the expected outcome, since it combines the strengths of all models.

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I. Introduction

As demand for electricity increases in both developed and developing countries, the global energy transition is accelerating. With the prospect of decarbonizing economies comes the rise of renewable energy sources (RES) such as wind, solar, hydropower, and geothermal energy [1]. The greatest increase is expected in wind and solar generation, together with measures for increased energy efficiency [1] [2]. One challenge that follows the transition to a large amount of RES is decreased security in terms of meeting electricity demand [3]. This is due to the high variability and uncertainty in the generation from sources like wind and solar energy, as they are dependent on variable factors like wind speed, cloud cover, and seasonal changes in sunlight. The intermittent nature of these weather-dependent energy sources makes them difficult to integrate into the power system [4]. To ensure safe grid operation, accurate forecasts of renewable energy generation are of vital importance. Other applications are within energy markets, grid stability, congestion management, storage optimization, and operational planning [3] [5].

Traditional forecasting tools often fail to capture the inherent uncertainty of weather-dependent generation. This includes the irregular relationship between solar radiation and weather factors such as cloud cover and atmospheric conditions and the non-linearity and randomness of wind conditions like wind speed and direction [4].

A large part of the UK energy strategy to increase energy independency and security is decarbonization. This includes investments into onshore and offshore wind and solar power [6]. The wind park Hornsea 1 is located 120 km off the coast of Yorkshire, England. It has an area of 407 square kilometers and a total capacity of approximately 1.2 GW. The project also includes the construction of larger wind farms in the same area [7].

Forecasting power production from renewable energy is a core part of Ørsted's business. As a result of that, IEEE Power & Energy Society Working Group on Energy Forecasting and Analytics sponsored by Ørsted and rebase.energy have organized a competition regarding power generation forecast of the Hornsea 1 and the combined solar generation of east England. The forecasts should be a day-ahead forecast for each half-hour period and should be in the form of quantiles with increments of 10% in the range of 10% to 90%. The quality of the forecasts will be determined using pinball loss [8].

The objective of the project is to produce competitive probabilistic forecasts of the combined wind and solar generation of Hornsea 1 and East England's solar capacity using different forecasting models. The aim is to train and test the models on the time period 2020-2023 and yield results that compete the results of previous competitors in several quantiles (10%-90%).

II. Renewable Energy Forecasting

Renewable energy forecasting plays a critical role in the integration of variable sources like solar and wind into power systems. In this section, different forecasting techniques, models and evaluation metrics are introduced.

A. Forecasting Techniques

Producing forecasts can be conducted either as point forecasts or probabilistic forecasts. In point forecasts, only a single value is estimated for each time step. Point estimates are limited in producing forecasts of features with uncertain nature, as they do not present the uncertainty in the estimation. In probabilistic forecasting, the spread of the distributions is estimated instead [3].

Probabilistic forecasting can be done either as quantile or density forecasts. The aim of quantile forecasts is to estimate the distribution of the data at each time step with no interdependencies, meaning that the distributions at each time step is independent from each other. In contrast to density forecasts that estimate the distribution as a continuous density function, the quantile forecasts estimate it as discrete quantiles. The quantiles define intervals that are equally spaced between levels of probability, for example 10, 20,..., 90% quantiles [3].

B. Different Types of Models

Traditionally, statistical models have been used to produce time series forecasting. Statistical models utilize some statistic function to produce an output of estimating some future data points using an input of historical data points. Different models use different mathematical algorithms to constitute this forecasting function. Some well-known statistical models include the ARCH model (Autoregressive Conditional Heteroskedasticity), the ARIMA (Autoregressive Integrated Moving Average) and the VAR model (Vector Autoregression) [9]. The main advantage of statistical models is simplicity in terms of implementation and interpretation [3].

A more recently developed field within forecasting is machine learning models. These utilize machine learning functions, and can handle non-linear relationships in the data. They are generally more expensive in terms of computation, but can handle larger and more complex datasets. Machine learning models also typically include measures to avoid overfitting the data, to make sure that the learning is more general and not tied to a specific dataset. Deep learning is a subset of machine learning that utilizes even more advanced learning algorithms [3].

C. Review of Models in the Competition

The results of the competition show that regression models gave the best forecasting results compared to other models. This is because these types of models effectively handles different data types, works well with smaller datasets (around 10k data points), is robust towards missing data, supports sharp decision boundaries and effectively filters out irrelevant data. The competition methods that produced the best results were CatBoost, XGBoost and LightGBM as well as multi-model ensembles [10].

D. Evaluation Metrics

The main evaluation metric that will be used for this report is the pinball loss but can also be referred to as the quantile loss, which can be describes as the following equations.

$$\begin{cases} y_\tau(y,\hat{y}) = \tau(y-\hat{y}) \text{ if y} \geq \hat{y} \\ y_\tau(y,\hat{y}) = (1-\tau)(\hat{y}-y) \text{ if } \hat{y} > \text{y} \end{cases} \tag{1}$$

Here, τ is the target quantile of the forecast which in this report is increments of 10% in the range of 10% to 90%. y is the real value and $\hat{y}$ is the predicted value of the quantile forecast. The pinball loss "penalize" predicted values that differ from the real value depending on the quantile used. The most important result of the pinball loss is that the lower the pinball loss, the better the prediction [11].

The MAE or the mean absolute error is one of the most common error measures. It is calculated by taking the average of the absolute errors over the entire time series [3].

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \tag{2}$$

The coefficient of determination or the R^2 value returns values between zero and one, where one is the best prediction which corresponds to a perfect fit for the observations. The model can also have a negative values, which indicate that the model is worse than if the mean value was used as the estimate.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \tag{3}$$

Here, $\bar{y}$ is the mean of the observations.

III. Data Description

Prior to modeling, the data is pre-processed and cleaned to ensure quality and consistency. In addition, exploratory data analysis is performed to identify patterns, trends, and key variables.

A. Data Pre-Processing

The raw meteorological data is provided in NetCDF format from the DWD ICON-EU forecasting model, for Hornsea 1 (wind) and PES10 (solar) regions. Each variable is a multidimensional array with reference datetime (initialization), valid datetime (lead time), and spatial dimensions latitude/longitude for wind and point indices for solar.

The reference datetime indicates the timestamp at which the forecast model was run, typically at 6-hour intervals, while the valid datetime represents the lead time in hours after initialization for which the forecast is valid. Each forecast run contained different time horizon predictions, resulting in overlapping forecast trajectories across time.

For the Hornsea 1 region, gridded forecasts covering latitudes from 53.77° to 54.10° and longitudes from 1.702° to 2.027° are spatially averaged across 36 grid points. Similarly, for PES10, solar forecasts from 20 spatially distributed measurement points are averaged to reduce complexity. This aggregation produces univariate time series for each variable, indexed by reference datetime (forecast initialization time) and valid datetime (forecast horizon in hours). The actual forecast timestamps are obtained by adding the forecast lead time to the initialization time and all times were converted to UTC for consistency.

Energy data includes wind and solar power generation in units of megawatts (MW) at half-hourly intervals. Since MW represents instantaneous power, it is converted to energy in megawatt-hours (MWh) by accounting for the 30-minute settlement period. Specifically for wind generation, the bid and offer acceptance (bOA) volumes, which represent curtailments and are reported in MWh, are subtracted to calculate the net energy volume credited to the market: Credit $=$ 0.5 (h) $\times$ Power (MW) $-$ bOA (MWh).

All available weather forecast features for the Hornsea 1 wind region and PES10 solar region are merged based on common forecast initialization and lead times. The merged weather dataset is resampled to a 30-minute resolution to match the settlement period of the energy market data. All forecasts are then joined with the observed energy generation values using the valid timestamps as keys. Since multiple forecasts may exist for the same valid time, each originating from a different initialization, only the most recent forecast available per timestamp is retained. This proposed design choice ensures that the dataset reflects realistic operational conditions, where only the latest and most reliable forecast would be used for energy prediction which is important for training models in a way that closely aligns with real-time forecasting deployment.

After pre-processing, the final dataset is split into training and test sets using a sequential, time-based approach. A temporal gap is introduced between the two sets to prevent potential data leakage. The training set includes all records prior to November 25, 2022, while the test set comprises data after December 6, 2022. This specific separation is chosen because the majority of missing values are concentrated around these two periods. Splitting the dataset early in the pipeline is crucial to avoid information leakage, especially in time-dependent data with high autocorrelation. Randomly splitting data can result in overly optimistic performance metrics. Instead, ensuring that the test set represents future data provides a more realistic evaluation of model performance [12]. Within the training set, an additional 80/20 time-based split is applied to form a validation set. This internal validation set is used for hyperparameter tuning and to evaluate models using grid search and cross-validation. This approach also enables the integration of the stacking ensemble method, where base learners are trained on the training subset and the meta-model is trained on predictions from the validation subset. Within the training set, missing values are addressed using linear interpolation, resulting in a complete dataset suitable for supervised learning. In contrast, missing values in the test set, particularly those in target variables, are left unchanged to maintain the reliability of model evaluation.

B. Data Analysis

A comprehensive exploratory data analysis (EDA) is conducted to uncover temporal patterns, seasonality, and correlations relevant to renewable energy production. The results of this analysis supports both informed model design and robust feature engineering and selection.

Monthly sums of wind and solar energy production are computed, revealing clear seasonal trends shown in Appendix B. Solar output peaks during spring and summer months, while wind production is more prominent during winter, in alignment with known climatological cycles in East England. To ensure the validity of monthly aggregation, September 2020 and November 2022 are excluded from visualizations and statistical summaries, as the dataset for these months did not cover full measurement periods.

To formalize these trends, binary seasonal indicators are added to the data set, denoting the high-solar (April to September) and high-wind (October to March) months. These labels are assigned based on whether monthly generation exceeds the overall monthly mean in more than two seasonal periods. Additionally, a daylight indicator is constructed using sunrise and sunset data specific to the East England region [13].

Further temporal features are derived through cyclical encoding of hour of day, day of week, and month of year. These encodings preserve the periodic nature of time-related variables, enabling learning models to capture daily and seasonal fluctuations [3].

Pearson correlation matrices are created for wind and solar features to examine their linear relationship with energy outputs. These can be reviewed in Appendix B. As expected, solar production is strongly correlated with radiation, and wind production with wind speed. While some features show low correlation with the targets, excluding them reduces model accuracy. Non-linear models can capture complex interactions and non-linear relationships that linear correlation measures might miss [3].

IV. Methodology

The forecasting framework aims to generate accurate probabilistic forecasts for both total solar and wind generation. Point forecasts are included as a complementary measure. The machine learning models used are Random Forest Regression from the scikit-learn library (BSD license) [14], Categorical Boosting from the catboost library (Apache 2.0 license) [15] and Light Gradient Boosting Machine from the lightgbm library (MIT license) [16], as well as an ensemble model that combines these models using scikit-learn [14]. The stacking methods are based on quantile regression for probabilistic forecasting and linear regression for point forecasting. These models were selected based on their strength of producing accurate forecasts with limited amount of data points as well as their ability to handle non-linear relationships.

A. Forecasting Framework

The original dataset is chronologically partitioned into training, validation, and test subsets. The training and validation sets are used for model development, feature selection, and hyperparameter tuning, while the test set is strictly reserved for the final performance evaluation of all models on unseen data. Within this framework, various combinations of lagged features from past target values and future covariates are explored and included. Lagged features are based on historical values of the target variables to capture temporal dependencies and autocorrelation. Future covariates are derived from numerical weather prediction (NWP) forecasts providing exogenous information about expected conditions. To improve generalization and reduce complexity, feature importance analysis for each model is performed during the training process in order to identify the most efficient predictors and exclude redundant or noisy variables. For each model multiple combinations of input features are evaluated in the modeling pipeline, with the objective of minimizing forecasting error under different configurations.

A critical component of the methodology is the hyperparameter optimization. Grid search combined with time series cross-validation is used for every base model to improve robustness.

Random Forest, CatBoost, and LightGBM are independently trained and evaluated using both deterministic and probabilistic evaluation metrics. A naive benchmark model is also implemented to serve as a reference baseline for both deterministic and probabilistic evaluations. This simplistic model replicates the value of the generation from the previous day as its predictions for future time steps.

B. Decision Tree Regression

All selected models utilize the concept of decision tree regression as their base learner. The general idea of a decision tree is to split the training observations into distinct and non-overlapping subsets to create tree-like structures. For each split in the decision tree, there is a corresponding piecewise function. These piecewise functions are of the following form [3].

$$f_p(\mathbf{X}) = \sum_{p=1}^{P} \alpha_p \chi_p(\mathbf{X}) \tag{4}$$

Here P denotes the number of subsets, while $\chi_p(\mathbf{X})$ is a characteristic function with $\mathbf{X}$ denoting the domain holding the observations. With C_p denoting the partition set containing N_p observations, this characteristic function can be described as [3]:

$$\chi_p(\mathbf{X}) = \begin{cases} 1 & \text{if } \mathbf{X} \text{ is in } \quad C_p \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

Lastly, α_p is an average of the dependent variables corresponding to independent variables in C_p and is formulated as [3]:

$$\alpha_p = \frac{1}{N_p} \sum_{t=1}^{N} L_t \chi_p(\mathbf{X}_t) \tag{6}$$

This piecewise function is used to represent the output of an individual regression tree, where each leaf node in the feature space corresponds to a prediction. The prediction for a new input $\mathbf{X}$ is obtained by identifying which partition it belongs to, and assigning the average value of the historical observations of the dependent variable in that partition [3].

C. Random Forest Regression

Random Forest Regression (RFR) is a widely used ensemble learning method based on decision tree regression, particularly effective for capturing non-linearities. It constructs a large number of individual regression trees, each trained on a randomly sampled subset of the training data using bootstrap sampling [17]. At each decision node, only a random subset of features is considered for splitting, which increases diversity among trees and helps prevent overfitting [17] [18].

By combining the predictions of different trees, the method reduces the variance and improves its generalization performance. Its robustness to noise and ability to model complex feature interactions make it suitable for forecasting tasks that involve a combination of meteorological, seasonal, and temporal variables [18].

To maximize the model's predictive performance, a set of important hyperparameters must be carefully tuned. These parameters control the structure and the behavior of the trees [3]:

- **n_estimators**: The number of trees in the forest. More trees generally lead to better performance but increase computational cost.
- **max_depth**: The maximum depth of each tree. Restricting depth helps prevent overfitting and improves the interpretability of the model.
- **min_samples_leaf**: The minimum number of samples required to be present in a leaf node. Higher values yield smoother predictions.

D. Light Gradient Boosting Machine

Gradient-Boosted Regression Trees (GBRT) are a set of machine learning algorithms included in Light Gradient-Boosted Regression Trees, but also in other popular models such as Extreme Gradient Boosting (XGBoost), and Categorical Boosting (CatBoost) [16]. In GBRT, the decision tree regression learners are trained in sequence, as compared to for example RFR where they are trained parallelly. In this way, the model can iteratively improve the current predictions based on the error of previous iterations, using the gradient of some loss function [3].

The main algorithm of a GBRT model is the weighted sum over M iterations of weak learner predictions, denoted $h_i(\mathbf{X})$ with weight γ_i [3].

$$f(\mathbf{X}) = \sum_{i=1}^{M} \gamma_i h_i(\mathbf{X}) \tag{7}$$

At each step, the model aims to improve the prior function by adding a new weak learner scaled by the learning rate α [3],

$$f_i(\mathbf{X}) = f_{i-1}(\mathbf{X}) + \alpha \gamma_i h_i(\mathbf{X}) \tag{8}$$

In the case of large amounts of data and features, GBRT models generally lack in efficiency and scalability. LightGBM was suggested as a method to avoid these issues by implementing two additional model features, Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). The GOSS feature is related to down sampling of data. Instead of random sampling, LightGBM prioritizes data instances with larger gradients as this is related to a larger contribution to the information gain. Instead it randomly drops data instances with small gradients, effectively assigning a weight for each data instance. The EFB feature bundles exclusive features to add efficiency and speed up computation [16]. Furthermore, LightGBM utilizes leaf-wise growth as compared to level-wise growth of the decision trees. This tends to achieve lower loss, but might lead to overfitting [19].

Important hyperparameters in GBRT are [3]:

- **n_estimators**: The number of trees. More trees generally lead to better performance but increase computational cost.
- **max_depth**: The maximum depth of each tree.
- **learning_rate**: A scaling factor determining the step size of each iteration

E. Categorical Boosting

Similarly to LightGBM discussed in Sec. IV-D, Categorical Boosting (CatBoost) uses Gradient-Boosted Regression Trees. The most prominent model feature of CatBoost is how it handles high cardinality categorical features where one-hot encoding is not sufficient to categorize it. For these instances, Ordered Target Statistic (Ordered TS) is used. The categorical values of a feature are replaced with a target statistic. A target statistic summarizes the relationship between the target variable and each category of a feature. Ordered TS ensures encoded categorical values using all of the information available in the training data without target leakage [15].

Furthermore, CatBoost uses a refinement of the GBRT technique referred to as Ordered Boosting that prevents prediction shift and target leakage by using random permutations of the dataset and only learning from the past data when training on a point. CatBoost also utilises an ensemble of Oblivious Decision Trees (ODTs) that are full binary trees, meaning that a tree with n levels has 2^n nodes. The ODTs are symmetric, which states that all non-leaf nodes use the same split condition at each node [15].

The CatBoost model is sensitive to hyperparameters and tuning them is of importance. For example, the usage of full binary ODTs means that with every unit of increase in the maximum depth, the memory use grows by twice the number of trees in the ensemble [15]. Important hyperparameters to tune are [3]:

- **iterations**: Corresponds to n_estimators or the number of trees.
- **max_depth**: The maximum depth of each tree.
- **learning_rate**:A scaling factor determining the step size of each iteration

F. Ensemble Model

To leverage the strengths of the individual machine learning models, a stacked ensemble approach is implemented. For each quantile level, the validation set predictions from the three base models Random Forest, CatBoost, and LightGBM are used as input features to train a meta-model via QuantileRegressor, with the pinball loss function guiding the optimization. This training process in the validation set identifies the optimal quantile weights that effectively combine the strengths of the base models to improve the accuracy of the forecast. Once the meta models are trained, they are applied to the corresponding test set predictions of the base models to generate final ensemble forecasts. This stacking based approach enables the ensemble to adapt better across target variable, leveraging different models depending on their strengths per quantile level. A similar procedure is implemented for point forecasting. Linear regression is used as the meta-learner, trained on the deterministic predictions of the three base models in the validation set. The meta-model is optimized using MAE and then applied to the test set, allowing for enhanced accuracy in single-valued predictions as well. Stacking has been shown to enhance both calibration and sharpness in probabilistic forecasting tasks, particularly in settings that involve heterogeneous base learners and distributional targets [20].

V. Results

The results are presented based on plots and error measures of the predictions of the test set across the solar, wind and combined generation of the ensemble model, that outperformed the other models. The corresponding results for the other models can be seen in Appendix C. The error measures of the predictions of the combined generations of the different models are then compared with each other and the naive model benchmark.

A. Ensemble Model

The combined predictions of solar and wind generations for all quantiles for the ensemble model compared to the actual value of the combined generation during the last two days of the test set is presented in Fig. 1. The different error measures for the wind, solar and combined predictions are presented in Table I.

Fig. 1. Combined Quantile Forecast for Solar and Wind Using the Ensemble Model

TABLE I
ERROR MEASURES FOR ENSEMBLE MODEL FORECASTS ACROSS SOLAR, WIND, AND COMBINED OUTPUTS

Ensemble model	MAE (MWh)	R^2	Pinball Score (MWh)
Solar Forecast	27.45	0.9278	10.45
Wind Forecast	63.30	0.8072	24.00
Combined Forecast	75.85	0.8439	29.39

B. Model Comparison

A table of the error measures for the combined predictions of wind and solar generations of all models compared to the naive model benchmark are presented in Table II. The best results are highlighted in bold symbols. Compared to the naive benchmark, the performance of the best model in the test set are significantly stronger with a reduction of around 60% in pinball score, and an increase in R^2 from 0.4594 to 0.8439.

VI. Discussion

The five most important features for the solar and wind predictions for the three models are listed in Tables VI-VII in Appendix D. For all models, solar radiation and wind speed

TABLE II
ERROR MEASURES FOR ALL FOUR MODELS COMPARED TO BENCHMARK

Train set	MAE (MWh)	R^2	Pinball score (MWh)
Random Forest	72.80	0.8635	30.57
LightGBM	131.29	0.6053	28.20
CatBoost	**68.30**	0.8677	27.47
Ensemble Model	68.80	**0.8729**	**26.71**
Naive Model	147.08	0.4408	73.54
Test set	MAE (MWh)	R^2	Pinball score (MWh)
Random Forest	79.40	0.8357	34.21
LightGBM	131.02	0.6042	30.45
CatBoost	**74.65**	0.8412	30.08
Ensemble Model	75.85	**0.8439**	**29.39**
Naive Model	146.28	0.4594	73.14

are the most important features for wind and solar forecasts respectively. This observation is supported by the correlation matrices, where these features show high correlation with generation and confirm their importance across all models.

All models perform with marginally better error measures than the naive benchmark as presented in Table II. This arguably shows that they are all at least a minimum viable product. Among the individual learners, CatBoost achieved the most competitive results as it performs the lowest MAE and highest R^2, as well as the lowest pinball score. LightGBM, while fast and efficient, has weaker performance as seen in Table II, and more specifically in wind prediction (this is seen in Appendix C). However LightGBM shows no signs of overfitting, as the evaluation on the train and test set show a similar result. This is in contrast to the other models that all have a performance drop indicating overfitting.

When comparing all models, the ensemble model achieve the best overall results, with the lowest pinball loss and the highest R^2 for combined forecasts. The stacking mechanism allow the meta-model to effectively blend the strengths of the base models and mitigating their individual weaknesses, enabling better generalization and quantile estimation.

The results of the top 15 competitors in the competition range between a pinball score of 22.18 - 33.92 MWh [8], meaning that the score of 29.39 MWh is arguably a competitive result. Furthermore, the models inhabit other strengths compared to the top competitors, as they are likely less costly and more transparent, as well as publicly accessible.

Although the ensemble model performs competitively relative to the competition, certain methodological enhancements could further improve the results. For example, CatBoost still achieves a lower MAE as seen in Table II. The weak link in the model is the LightGBM model. However, since it does not perform well on the validation set, it will not receive significant weight in the ensemble. As the weight of each model is assigned from the performance on the train set, the LightGBM predictions might not have much influence on the results of the meta model. Even though the poor performance of LightGBM might not have significant negative impact, tuning the LightGBM model could still positively improve the performance of the ensemble model by adding more input and robustness. Further improvements like clipping quantile outputs to predefined physical capacity boundaries could also enhance the performance.

VII. CONCLUSION

The CatBoost model performed best among individual models, while the ensemble model achieved the strongest overall results. Its competitive pinball score suggests the aim to match IEEE competition results was met. The model could however be further improved through improving the individual models or adding capacity boundaries. The results are also argued to have strengths in terms of accessibility, transparency and affordability.

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APPENDIX A
GITHUB REPOSITORY

GitHub Repository – EG2140 Project Group 8 (MIT license)

APPENDIX B
DATA ANALYSIS

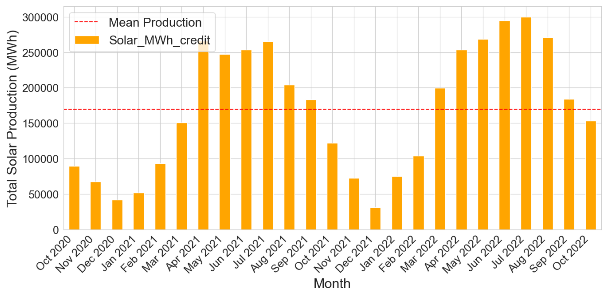

Fig. 2. Monthly Solar Energy Production

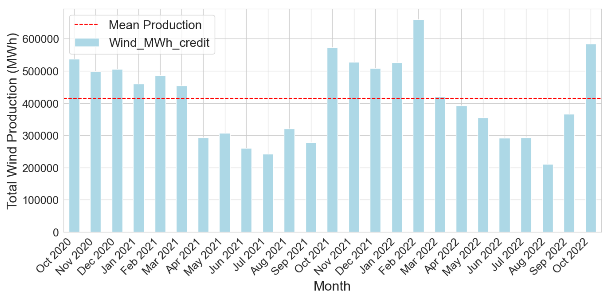

Fig. 3. Monthly Wind Energy Production

Fig. 4. Solar Feature Correlation Matrix

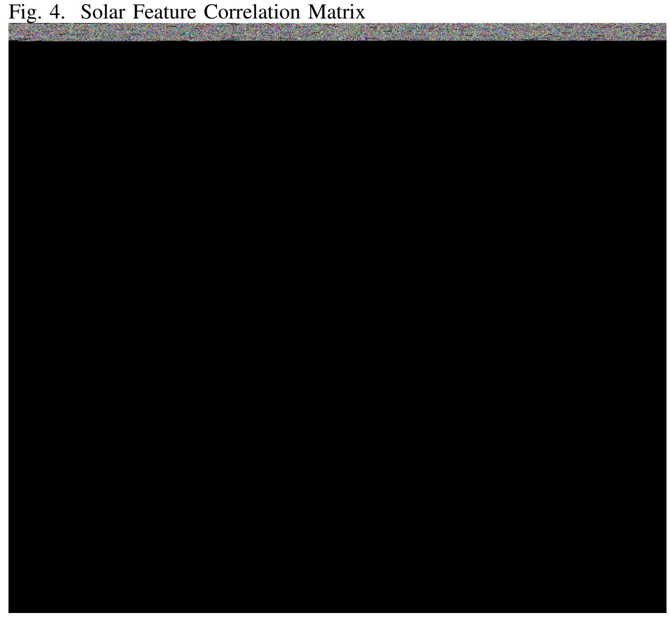

Fig. 5. Wind Feature Correlation Matrix

Appendix C
Additional Results

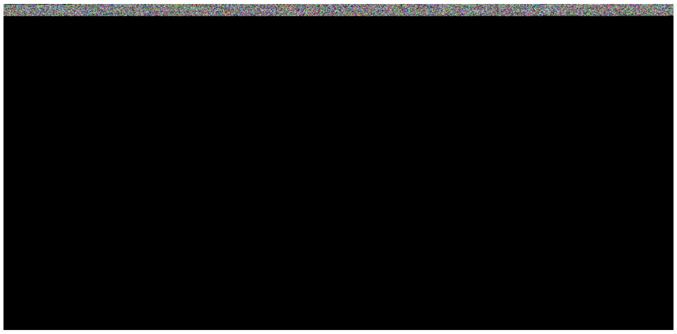

Fig. 6. Combined Quantile Forecast for Solar and Wind Using Random Forest Regressor

Fig. 7. Combined Quantile Forecast for Solar and Wind Using CatBoost

Fig. 8. Combined Quantile Forecast for Solar and Wind Using LightGBM

TABLE III
Error Measures for Random Forest Forecasts across Solar, Wind, and Combined Outputs

Train set	MAE (MWh)	R^2	Pinball score (MWh)
Solar Forecast	23.78	0.9435	10.29
Wind Forecast	62.54	0.8268	26.58
Combined Forecast	72.80	0.8635	30.57
Test set	MAE (MWh)	R^2	Pinball score (MWh)
Solar Forecast	27.24	0.9234	11.98
Wind Forecast	66.50	0.7987	29.00
Combined Forecast	79.40	0.8357	34.21

TABLE IV
Error Measures for LightGBM Forecasts across Solar, Wind, and Combined Outputs

Train set	MAE (MWh)	R^2	Pinball Score (MWh)
Solar Forecast	41.11	0.8648	9.45
Wind Forecast	96.21	0.6229	23.56
Combined Forecast	131.29	0.6053	28.20
Test set	MAE (MWh)	R^2	Pinball Score (MWh)
Solar Forecast	37.60	0.8686	10.52
Wind Forecast	100.72	0.5877	25.12
Combined Forecast	131.02	0.6042	30.45

TABLE V
Error Measures for CatBoost Forecasts across Solar, Wind, and Combined Outputs

Train set	MAE (MWh)	R^2	Pinball Score (MWh)
Solar Forecast	24.38	0.9398	9.59
Wind Forecast	56.90	0.8346	22.50
Combined Forecast	68.30	0.8677	27.47
Test set	MAE (MWh)	R^2	Pinball Score (MWh)
Solar Forecast	26.62	0.9265	10.54
Wind Forecast	60.97	0.8050	24.54
Combined Forecast	74.65	0.8412	30.08

Appendix D
Feature importance

TABLE VI
Feature Importance for the Solar and Wind Random Forest Regression Model Respectively

Rank	Feature	Importance
1	SolarDownwardRadiation	0.9498
2	Hour_cos	0.0211
3	CloudCover	0.0057
4	Temperature_y	0.0044
5	Solar_Mwh_lag48	0.0034
1	WindSpeed_100	0.9620
2	Temperature_x	0.0058
3	WindSpeed	0.0048
4	WindDirection_100_cos	0.0022
5	Day_sin	0.0022

TABLE VII
Feature Importance for the Solar and Wind LightGBM Model Respectively

Rank	Feature	Importance
1	SolarDownwardRadiation	0.2000
2	CloudCover	0.1317
3	Temperature_y	0.1290
4	Solar_MWh_lag48	0.0953
5	Hour_of_day	0.0793
1	Temperature_x	0.0993
2	WindSpeed_100	0.0824
3	WindSpeed	0.0587
4	WindSpeed_t24	0.0502
5	Wind_MWh_lag48	0.0458

TABLE VIII
Feature Importance for the Solar and Wind CatBoost Model Respectively

Rank	Feature	Importance
1	SolarDownwardRadiation	0.8518
2	Hour_cos	0.0613
3	CloudCover	0.0224
4	is_daylight	0.0194
5	Hour_sin	0.0124
1	WindSpeed_100	0.9669
2	Temperature_x	0.0064
3	WindSpeed	0.0058
4	WindDirection_cos	0.0045
5	WindDirection	0.0028